sen subgoal into a smooth, dense trajectory.

In this paper, we introduce **H**ierarchical **D**iffusion-**F**low (`HDFlow`), a new hierarchical framework for long-horizon planning that is built on this core insight. `HDFlow` leverages a hybrid generative architecture that assigns the right tool to the right job. For high-level planning, we employ a *diffusion model* to generate diverse sequences of subgoals within a learned latent space of a world model. While standard conditional diffusion models can generate plans that are consistent with the goal, they lack an explicit mechanism to assess the quality or long-term viability of those plans. In complex, sparse-reward settings, many plausible-looking sequences of subgoals can lead to irreversible failures. To address this, we introduce an *energy-based model* (EBM) to provide explicit guidance. The EBM is trained to assign low energy to successful strategies and high energy to failing ones, effectively learning a dense reward signal. This energy function then steers the diffusion planner away from potential dead ends and towards high-quality solutions, which is critical for robust, long-horizon performance. Furthermore, to mitigate the failures caused by inaccurate guidance, we enhance the standard EBM guidance with a two-step *manifold-aware* process (He et al., 2024). For low-level planning, we introduce a *rectified flow model* to rapidly generate dense latent trajectories to reach each subgoal.

We evaluate our proposed method on four contact-rich assembly tasks from the FurnitureBench (Heo et al., 2025) benchmark in both simulation and real-world settings. Furthermore, we evaluate HDFlow on two diverse and challenging benchmarks: RLBench (James et al., 2020) and OGBench (Park et al., 2025), allowing for a more comprehensive assessment of our method's robustness and generalization capabilities.

In summary, our contributions are as follows:

- We propose `HDFlow`, a novel, hybrid hierarchical planner that combines a diffusion model for high-level exploration and a rectified flow model for low-level trajectory generation.
- We introduce a two-stage training process featuring a contrastive-trained world model for structured representation learning and a manifold-aware EBM for explicit guidance of the high-level planner, enabling robust planning in sparse-reward environments.
- We demonstrate superior performance on various long-horizon benchmarks in both simulation and real-world settings.

2. Related Works

Recent advancements in generative models, particularly denoising diffusion probabilistic models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022), have significantly impacted planning in robotics by framing it as a conditional generation problem. Early works such as `Diffuser` (Janner et al., 2022) leverage iterative denoising to generate flexible trajectories conditioned on objectives like rewards or constraints. Building on this, Decision Diffuser (`DD`)(Ajay et al., 2023) further demonstrated that return-conditional diffusion models can outperform traditional offline reinforcement learning by enabling conditioning on various factors like constraints and skills. While powerful, the iterative nature of diffusion models can be computationally intensive (Dong et al., 2024). To tackle long-horizon tasks, hierarchical planning with diffusion models has emerged, as seen in (Li et al., 2023; Chen et al., 2024; Hao et al., 2025). Simple Hierarchical Diffuser (`SHD`)(Chen et al., 2024) employs a high-level diffusion model for sparse subgoal generation and a low-level one for dense trajectory refinement, aiming for improved efficiency and generalization. However, their reliance on diffusion models for both high-level and low-level planning increases the maintenance burden and makes them computationally expensive.

3. Preliminaries

3.1. Denoising Diffusion Models

Denoising diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020) are generative models that learn a data distribution $p(\mathbf{x})$ by reversing a fixed forward noising process.

Forward Process. The forward process gradually adds Gaussian noise to a data sample $\mathbf{x}_0$ over L discrete timesteps, according to a variance schedule β_ℓ: $q(\mathbf{x}_\ell|\mathbf{x}_{\ell-1}) = \mathcal{N}(\mathbf{x}_\ell; \sqrt{1-\beta_\ell}\mathbf{x}_{\ell-1}, \beta_\ell\mathbf{I})$. A key property is that we can sample $\mathbf{x}_\ell$ at any timestep ℓ in closed form: $q(\mathbf{x}_\ell|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_\ell; \sqrt{\bar{\alpha}_\ell}\mathbf{x}_0, (1-\bar{\alpha}_\ell)\mathbf{I})$, where $\alpha_\ell = 1-\beta_\ell$ and $\bar{\alpha}_\ell = \prod_{i=1}^{\ell}\alpha_i$. As $\ell \to L$, $\mathbf{x}_L$ approaches an isotropic Gaussian distribution $\mathcal{N}(0, \mathbf{I})$.

Reverse Process. The reverse process is a learned generative model that starts from noise $\mathbf{x}_L \sim \mathcal{N}(0, \mathbf{I})$ and iteratively denoises it to produce a sample. This process is modeled by a neural network $\epsilon_\theta(\mathbf{x}_\ell, \ell)$ trained to predict the noise ϵ that was added to the original sample $\mathbf{x}_0$ to produce $\mathbf{x}_\ell$. The training objective is to minimize the mean squared error between the true and predicted noise:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{\ell,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_\ell}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_\ell}\epsilon, \ell)\right\|^2\right] \tag{1}$$

For conditional generation (e.g., on a context c), classifier-free guidance (CFG) (Ho & Salimans, 2022) is commonly used. The model is trained on both conditional and unconditional inputs, and the noise prediction during inference is modified to steer generation towards the context: $\hat{\epsilon}_\theta(\mathbf{x}_\ell, \ell, c) = \epsilon_\theta(\mathbf{x}_\ell, \ell, \emptyset) + w \cdot (\epsilon_\theta(\mathbf{x}_\ell, \ell, c) - \epsilon_\theta(\mathbf{x}_\ell, \ell, \emptyset))$